

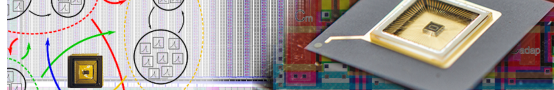


university of
groningen

faculty of science
and engineering

bio-inspired
circuits and systems

cognigron



A Title

RUG beamer-based presentations

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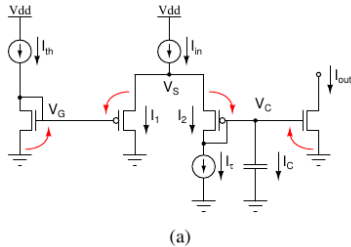
13/04/2026



set up this repository for all of my thesis work. going to attempt to set up obsidian maybe as well.

Translinear Principle

- › $\sum_{n \in CW} V_n = \sum_{n \in CCW} V_n$
- › $\prod_{n \in CW} I_n = \prod_{n \in CCW} I_n$
- › Valid for BJT's and MOSFET under weak inversion regime.



inversion regime's

- › weak inversion: sparse inversion layer???, diffusion transport
- › moderate inversion:
- › strong inversion: well formed channel????, drift transport

Log domain circuits

- › circuits that can be described by linear differential equations despite having nonlinear internal components
- › why are they called "Log domain"???

Technical questions

- › Latex(VScode) how to interrupt compiling operation if system is hung (stuck on `wpmain.synctex(busy)`) typically?

Admin questions

- › should I upload pdf's of all of my scanned notes that I have taken thusfar to this repository so that it is available?
- › could I send you a copy of the presentation I make for the group meeting thursday before I present?

random theory questions

- › If I understand this correctly when you refer to "maintaining the linear dynamics" as we go from n dynapses to 1 dynapse, that is the system is still "log domain" assuming I understood that principal correctly?

Goals for friday 17th (and Thursday 16th)

- › prepare presentation for the thursday meeting
- › read chapter 5 VLSI and take notes on it
- › read "Lab V - DPI Synapse and Axon-Hillock Neuron" write down list of questions.
- › schedule meeting with masters student in Zurich (send message after this meeting)
- › get in contact with the other BSc student?
- › priority reading list?? get through the 3 most important papers (if possible more)
- › recreate the circuits provided to me